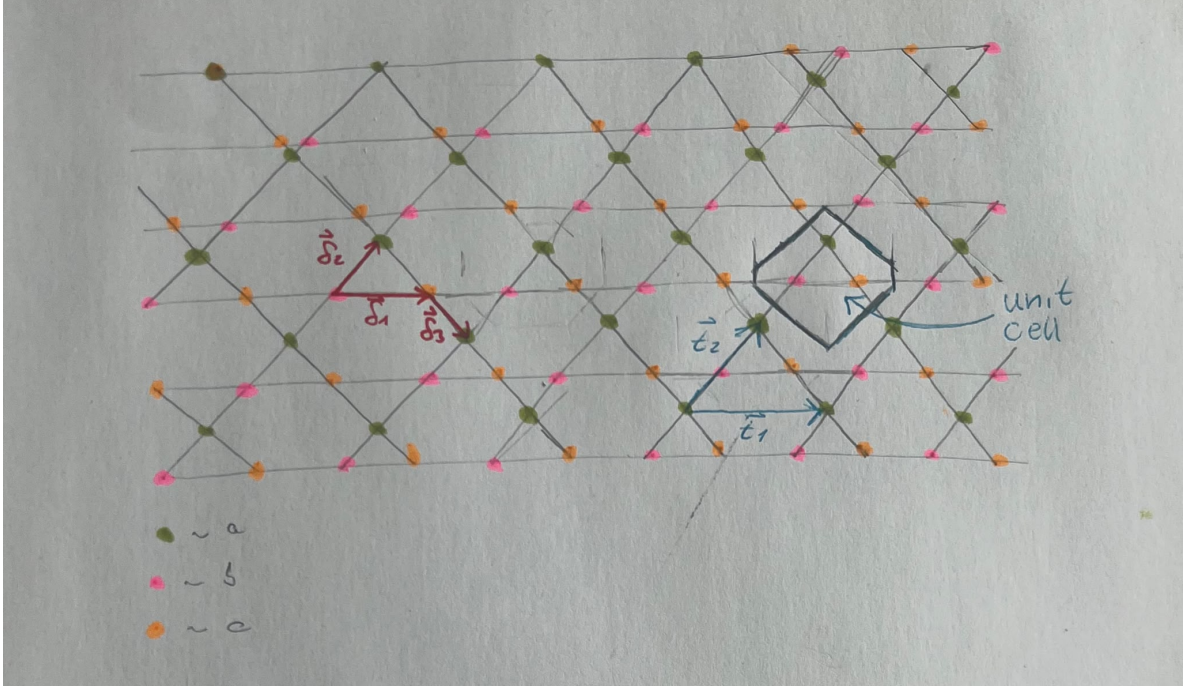


0.1 0-flux State

0.1.1 real space

The bare Kagome lattice is a triangular Bravais lattice with a three site unit cell (called a, b and c-sites below). The Wigner Seitz-cell is a hexagon (with full symmetries, even if the drawing suggests otherwise ;)). Geometrically speaking the lattice consists of corner-sharing equilateral triangles.



In all that follows I normalize all distances wrt the distance between nearest neighbours. The nearest neighbour vectors are (see cartoon)

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} \quad (1)$$

and the primitive translations

$$\vec{t}_1 = 2\vec{\delta}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = 2\vec{\delta}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}. \quad (2)$$

The tight binding hamiltonian is

$$H = t \sum_{i\sigma} \left(a_{i\sigma}^\dagger c_{i\sigma} + c_{i\sigma}^\dagger b_{i\sigma} + b_{i\sigma}^\dagger a_{i\sigma} + c_{i\sigma}^\dagger b_{i+t_1;\sigma} + c_{i\sigma}^\dagger a_{i+t_1-t_2;\sigma} + a_{i\sigma}^\dagger b_{i+t_2;\sigma} + \text{h.c.} \right) \quad (3)$$

The first index refers to the unit cell and the second one to the "species" of the particle, the latter being a product of the parton decomposition.

0.1.2 reciprocal space

The reciprocal lattice vectors are

$$\vec{b}_1 = \pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = \pi \begin{pmatrix} 0 \\ 2/\sqrt{3} \end{pmatrix} \quad (4)$$

The first Brillouin zone is also a hexagon (rotated by 90° wrt to the Wigner-Seitz cell). To find H in reciprocal space I do a Fourier transform:

$$a_{i\sigma} = \sqrt{\frac{3}{N}} \sum_k e^{i\vec{k}\cdot\vec{R}_i} a_{k\sigma}, \quad (5)$$

where N is the number of unit cells. This yields

$$H = 2t \sum_{q\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger & c_{q\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 0 & \cos(\vec{q} \cdot \vec{\delta}_2) & \cos(\vec{q} \cdot \vec{\delta}_3) \\ \cos(\vec{q} \cdot \vec{\delta}_2) & 0 & \cos(\vec{q} \cdot \vec{\delta}_1) \\ \cos(\vec{q} \cdot \vec{\delta}_3) & \cos(\vec{q} \cdot \vec{\delta}_1) & 0 \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \end{pmatrix}. \quad (6)$$

0.1.3 dispersion and band touching points

The three eigenvalues are

$$\varepsilon_1(\vec{q}) = t \left(1 - \sqrt{4 \cos^2(q_x) + 4 \cos(q_x) \cos(\sqrt{3}q_y) + 1} \right) \quad (7)$$

$$\varepsilon_2(\vec{q}) = t \left(1 + \sqrt{4 \cos^2(q_x) + 4 \cos(q_x) \cos(\sqrt{3}q_y) + 1} \right) \quad (8)$$

$$\varepsilon_3(\vec{q}) = -2t, \quad (9)$$

see [1]. I also used

$$\cos^2 \left(\frac{1}{2} (x + \sqrt{3}q_y) \right) + \cos^2 \left(\frac{1}{2} (q_x - \sqrt{3}q_y) \right) = \cos(q_x) \cos(\sqrt{3}q_y) + 1 \quad (10)$$

Interestingly, the top band is completely flat. The other two bands touch when

$$\cos^2(x) + \cos(x) \cos(\sqrt{3}y) + 1/4 = 0 \quad (11)$$

$$\Rightarrow \cos(q_x) = \frac{1}{2} \left(-\cos(\sqrt{3}q_y) \pm \sqrt{\cos^2(\sqrt{3}q_y) - 1} \right) \quad (12)$$

$$\Rightarrow q_y = \frac{\pi l}{\sqrt{3}}, \quad l \in \mathbb{N} \quad (13)$$

$$\Rightarrow q_x = (-1)^s \left(\frac{\pi}{2} + \frac{(-1)^l \pi}{6} \right) + 2\pi m, \quad m \in \mathbb{N}, \quad s \in [0, 1] \quad (14)$$

These touching points lie at the corners of the Brillouin zone, there are two "species". One can imagine that two of the parameters represent translation by reciprocal lattice vectors while the third one selects the species of the touching points. This not what m , l and s do but maybe one can transform them. Also, why does the solution depend on three parameters? For example

$$\vec{K} = \begin{pmatrix} 2\pi/3 \\ 0 \end{pmatrix}, \quad m = 0, s = 0, l = 0 \quad (15)$$

$$\vec{K}' = \begin{pmatrix} -2\pi/3 \\ 0 \end{pmatrix}, \quad m = 0, s = 1, l = 0. \quad (16)$$

0.1.4 filling factor

The idea is the following: The lower two bands contain the same number of states by symmetry. By calculating the number of states of the upper band (should be easy) it should be possible to calculate at what filling factor the low energy physics happen exactly the linear touching points.

0.1.5 linearization in reciprocal space

Use

$$\cos\left(\left(\vec{K}^{(\prime)} + \vec{p}\right) \cdot \vec{\delta}_{ij}\right) \approx \cos\left(\vec{K}^{(\prime)} \cdot \vec{\delta}_{ij}\right) - \sin\left(\vec{K} \cdot \vec{\delta}_{ij}\right) \left(\vec{p} \cdot \vec{\delta}_{ij}\right) \quad (17)$$

and

$$\begin{aligned} \vec{K} \cdot \vec{\delta}_1 &= \frac{2\pi}{3} \\ \vec{K} \cdot \vec{\delta}_2 &= \frac{\pi}{3} \\ \vec{K} \cdot \vec{\delta}_3 &= \frac{\pi}{3} \\ \vec{K}' &= -\vec{K} \end{aligned} \quad (18)$$

$$H_{\vec{K}} \approx t \sum_{p\sigma} (a_{p\sigma}^\dagger b_{p\sigma}^\dagger c_{p\sigma}^\dagger) \begin{pmatrix} 0 & 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_2 & 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_3 \\ 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_2 & 0 & -1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_1 \\ 1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_3 & -1 - \sqrt{3}\vec{p} \cdot \vec{\delta}_1 & 0 \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \end{pmatrix} \quad (19)$$

Directly at the $\vec{K}$ -point the eigenvalues are obviously

$$\varepsilon_1(\vec{K}) = t, \quad \varepsilon_2(\vec{K}) = t, \quad \varepsilon_3(\vec{K}) = -2t, \quad (20)$$

the eigenvector corresponding to the flat band reads at the $\vec{K}$ -point:

$$\vec{v}_3(\vec{K}) = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}. \quad (21)$$

Since H is hermitian (/symmetric in this case) there exist a orthonormal basis of eigenvectors. Obviously the subspace orthonormal to $\vec{v}_3(\vec{K})$ corresponds to the two bands that cross. A possible (orthonormal) basis for this subspace is

$$\vec{v}_1(\vec{K}) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{v}_2(\vec{K}) = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}. \quad (22)$$

Is it important to consider the phases of the states here?

Projecting H onto this basis yields

$$\begin{aligned} \vec{v}_1^T \cdot H_{\vec{K}} \cdot \vec{v}_1 &= t - \sqrt{3}t\vec{\delta}_2 \cdot \vec{p} & \vec{v}_1^T \cdot H_{\vec{K}} \cdot \vec{v}_2 &= -t(\vec{\delta}_1 + \vec{\delta}_3) \cdot \vec{p} \\ \vec{v}_2^T \cdot H_{\vec{K}} \cdot \vec{v}_1 &= -t(\vec{\delta}_1 + \vec{\delta}_3) \cdot \vec{p} & \vec{v}_2^T \cdot H_{\vec{K}} \cdot \vec{v}_2 &= t + \sqrt{3}t\vec{\delta}_2 \cdot \vec{p} \\ & \Leftrightarrow t\sigma_0 - \sqrt{3}t(\vec{\delta}_2 \cdot \vec{p})\sigma_z + t((\vec{\delta}_1 + \vec{\delta}_3) \cdot \vec{p})\sigma_x \end{aligned} \quad (23)$$

This already has the correct dispersion

$$\varepsilon_{1,2}(\vec{K} + \vec{p}) \approx \sqrt{3}t|\vec{p}|, \quad (24)$$

but is not yet in the form of the Weyl equation. **This can probably be achieved by rotating the basis in the degenerate subspace?**

Sanity check; directly linearizing the algebraic expressions for $\varepsilon_{1,2}(\vec{q})$ gives

$$\textit{Please work} \quad (25)$$

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